Image reading and display

1. Image reading:

img = cv2.imread('yahboom.jpg', 0) The first parameter is the path of the image, and the second parameter is how to read the image.

cv2.IMREAD_UNCHANGED: Keep the original format unchanged, -1;

cv2.IMREAD_GRAYSCALE: Read the image in grayscale mode, which can be represented by 0;

cv2.IMREAD_COLOR: Read a color image, which can be represented by 1; the default value

cv2.IMREAD_UNCHANGED: Read an image and include its alpha channel, which can be represented by 2.

2. Image display

cv.imshow('frame', frame): Open a window named frame and display frame data (image/video data)

Parameter meaning:

The first parameter indicates the name of the window to be opened

The second parameter indicates the image to be displayed

2.1. Code and actual effect display

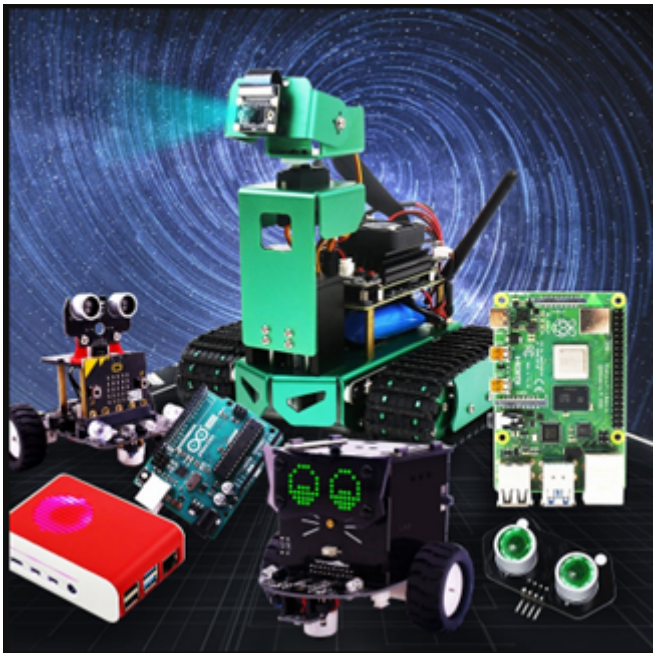
Code path:

```
~/dofbot_ws/src/dofbot_opencv/scripts/1.OpenCV_basic/01_read_pic.ipynb
```

Main code:

```
import cv2
img = cv2.imread('yahboom.jpg', 1)
# cv2.imshow('image', img) #This line can only be executed in the command line py
file, and a video window will pop up
# cv2.waitKey (0)
```

```
#bgr8 to jpeg format
import enum
import cv2
def bgr8_to_jpeg(value, quality=75):
return bytes(cv2.imencode('.jpg', value)[1])
```

```
# Image component in jupyterLab displays the read image
import ipywidgets.widgets as widgets
image_widget = widgets.Image(format='jpg', width=800, height=800)
display(image_widget)
image_widget.value = bgr8_to_jpeg(img)
```

After running the code block, you can see the following interface, the image has been read